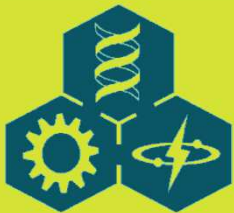
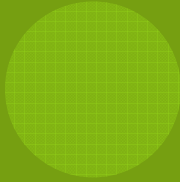


LECTURE 3:

QUALITATIVE MODEL FORMULATION





OUTLINE

3.1 How to Eat an Elephant

3.2 Forrester Diagrams

3.3 Examples

3.4 Errors in Forrester Diagrams

3.5 Advantages and Disadvantages of Forrester
Diagrams

3.6 Principles of Qualitative Formulation

3.7 Model Simplification

3.8 Other Modeling Problems



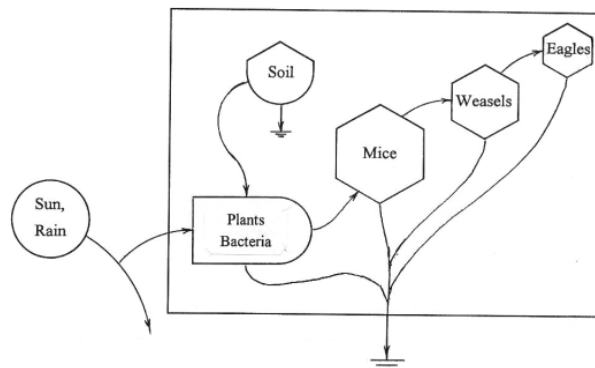
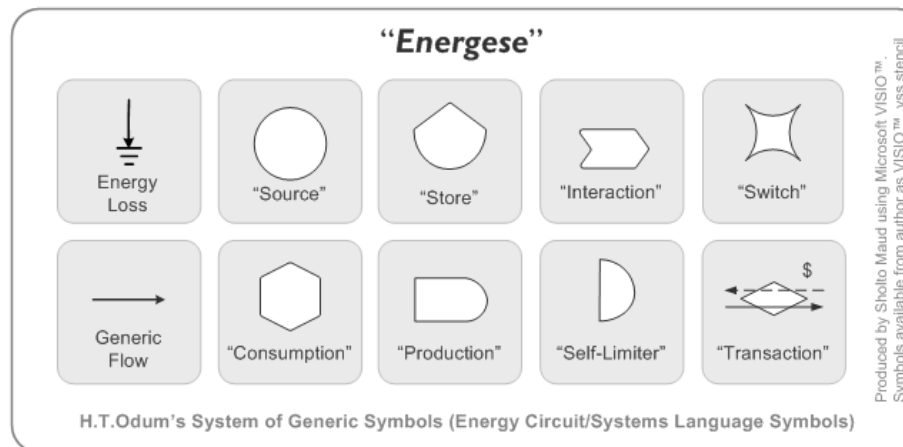
3.1 HOW TO EAT AN ELEPHANT

- ◎ Qualitative model formulation is the conversion of an objective statement and a set of hypotheses and assumptions into an informal, conceptual model.
- ◎ This form does not contain explicit equations, but its purpose is **to provide enough detail and structure so that a consistent set of equations can be written.**
- ◎ Qualitative models can take any form (except mathematical), but diagrams are the usual representation. Three important diagrammatic schemes are:
 - (1) Block structure diagram
 - (2) Odum energy flow diagram
 - (3) Forrester diagram

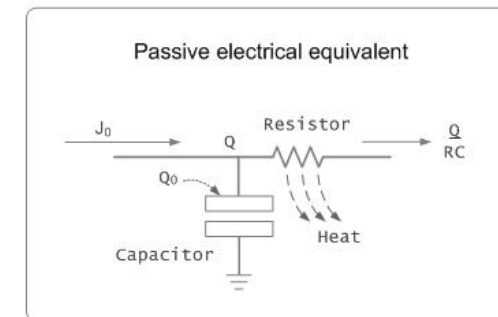
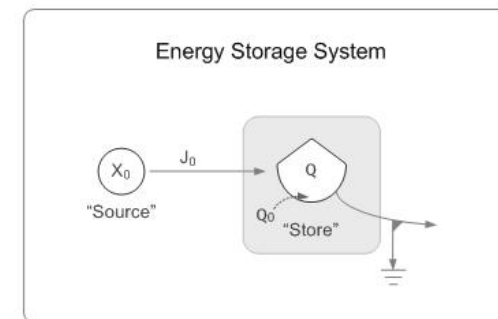


3.1 HOW TO EAT AN ELEPHANT

Odum energy flow diagram



An example of food chain



Adapted from H.T.Odum (1994) Fig. 3-8, p. 35

Source: http://en.wikipedia.org/wiki/Howard_T._Odum
<https://racerocks.ca/energy-diagrams/>



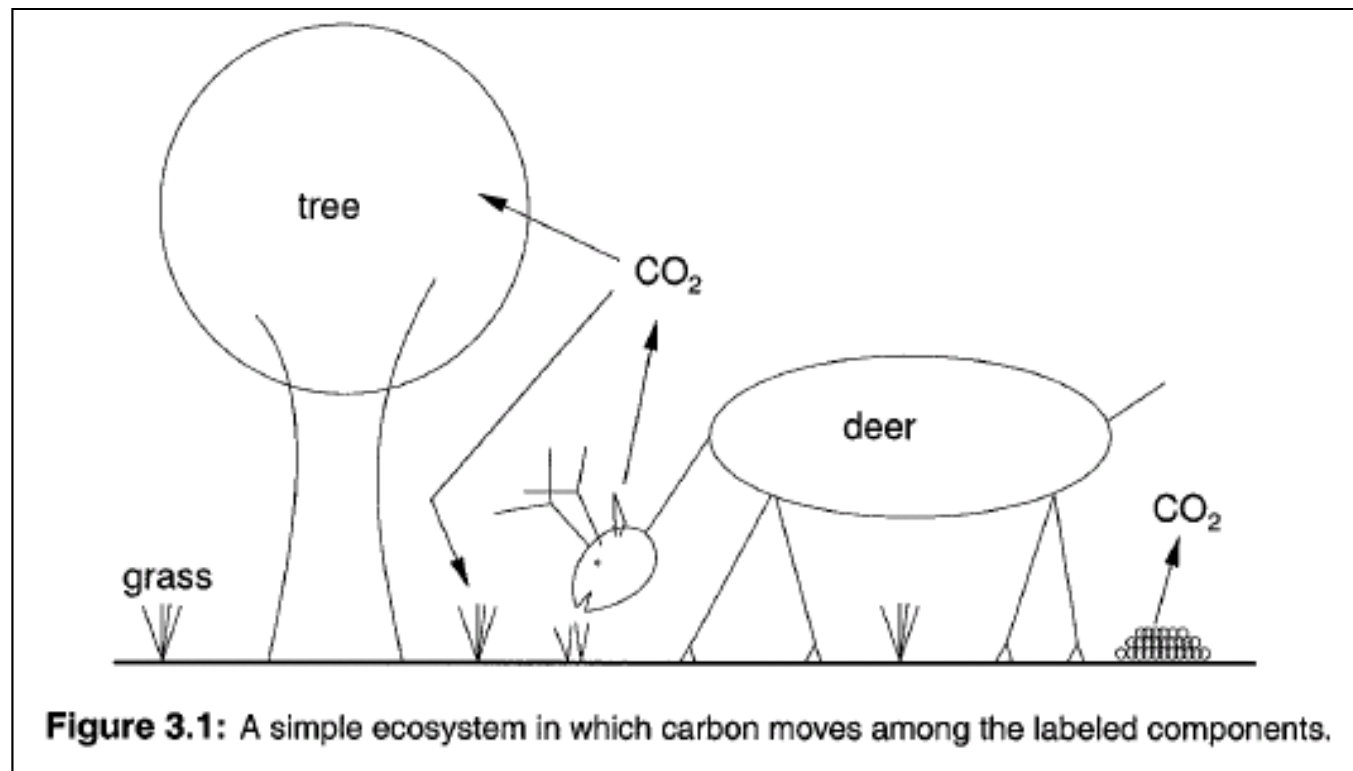
3.2 FORRESTER DIAGRAMS

- ◎ **Forrester diagrams** are designed to represent any dynamic system in which a measurable quantity flows between system components.
- ◎ Forrester diagrams are such an abstraction of the basic concepts of *system components* and *material flows* to obtain a general tool for qualitative modeling of systems.



3.2 FORRESTER DIAGRAMS

◎ A Simple Grass-Deer Ecosystem



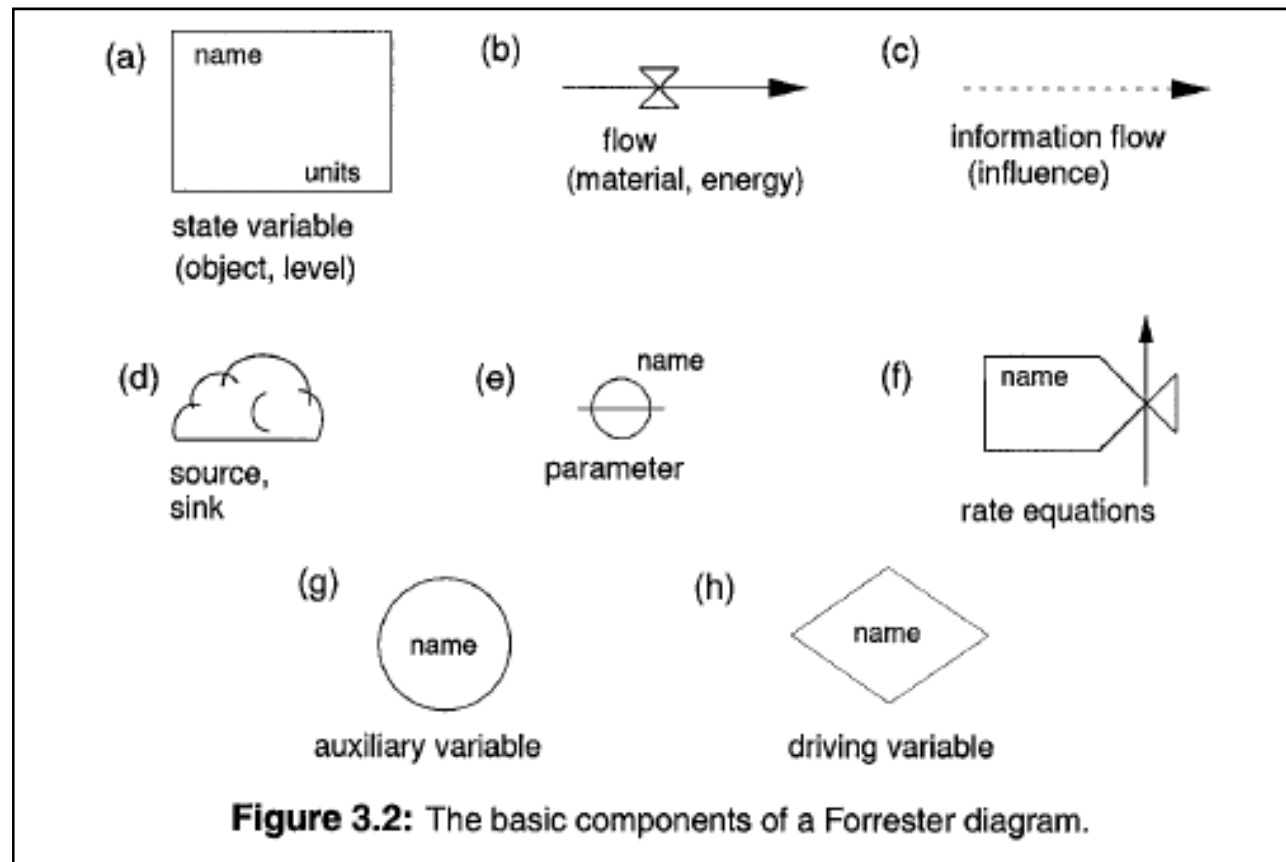
3.2 FORRESTER DIAGRAMS

- ◎ The *internal objects* that are modeled explicitly are called **state variables** and are those that, taken all together, characterize the condition or state of the system.
- ◎ The outside or *external variables* are either **sources** or **sinks** and are not modeled explicitly (i.e., no equations are written for these).
- ◎ *Relations* between system objects have two forms: (1) the direction and rates of flow between the quantity of interest and the objects and (2) the influences of a variable (e.g., the quantity of interest) on the rates of flow.



3.2 FORRESTER DIAGRAMS

◎ The Basic Components of a Forrester Diagram



3.2 FORRESTER DIAGRAMS

- ◎ **Objects:** They are the primary system components whose values over time we wish to predict.
- ◎ **Material Flows:** A flow is represented as a solid arrow and identifies the pathway over which the quantity of interest (e.g., grams of carbon) flows.
- ◎ **Information Flow or Influences:** The second manifestation of relations between objects are the effects that the quantity of one object has on the rates of inputs to or outputs from another object (e.g., effects on growth rates).



3.2 FORRESTER DIAGRAMS

- ◎ **Sources and Sinks:** Objects that are defined to be outside the system of interest, but which are inputs to state variables or outputs from state variables.
- ◎ **Parameters:** Constants in equations.
- ◎ **Rate Equations:** Total (or absolute) rates of input to, or output from, a state variable are described mathematically with rate equations.
- ◎ **Auxiliary Variables and Equations:** Auxiliary variables are variables that are computed from an auxiliary equation.
- ◎ **Driving Variables:** Dynamic events that relate to variables that are not state variables (e.g., season or temperature in some models) are often used as forcing functions.



3.3 EXAMPLES

◎ Forrester Diagram for the Grass-Deer Ecosystem

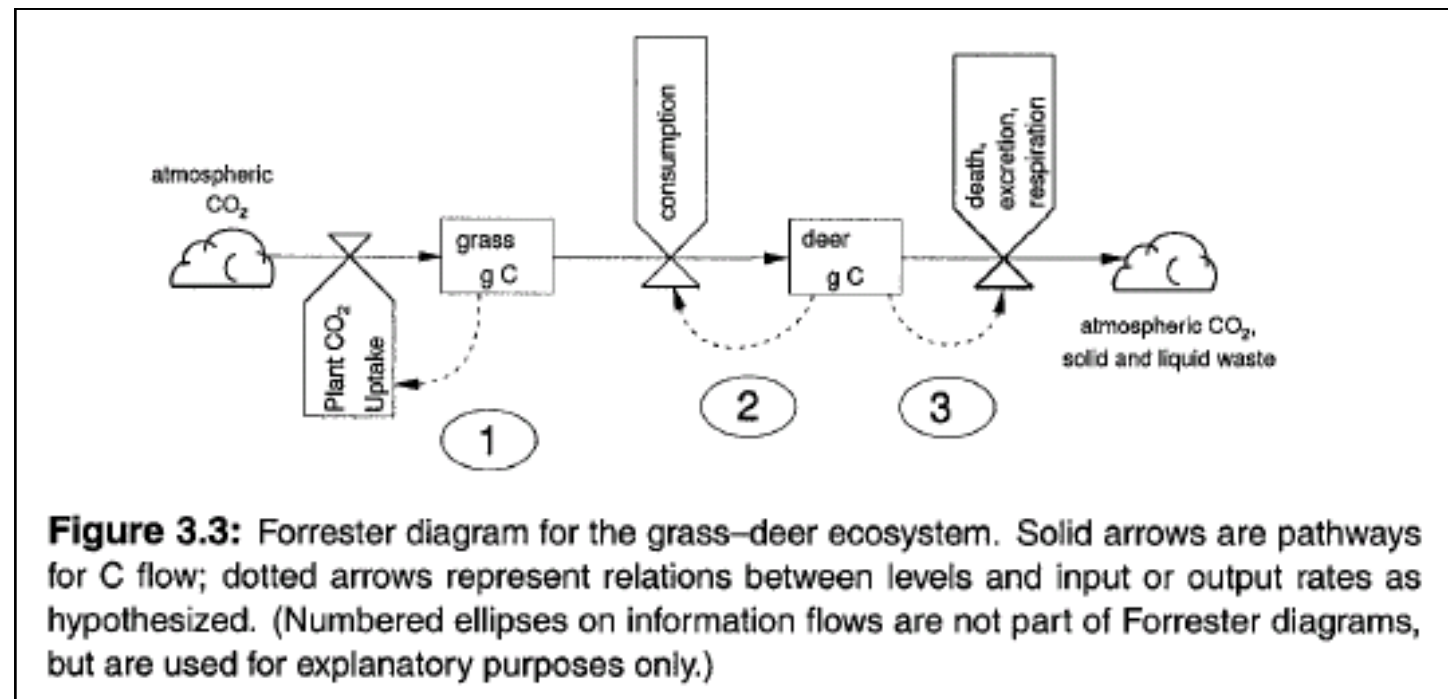
Assumptions:

1. The per capita rate of growth of grass (g C produced per g C of existing grass) is constant. Therefore, the total growth will be the per capita rate times the total amount of C present.
2. The only loss to the quantity of C in the grass population is by deer consumption.
3. Deer compete with one another for grass so that, as the quantity of deer increases, each deer receives less C.
4. Deer excrete or respire a fixed proportion of their existing C as either atmospheric C or solid/liquid waste.



3.3 EXAMPLES

☉ Forrester Diagram for the Grass-Deer Ecosystem



3.3 EXAMPLES

- Population Growth with Explicit Birth and Death
Density-Independent Model

$$N_{t+1} = N_t + bN_t - dN_t \quad (3.1)$$

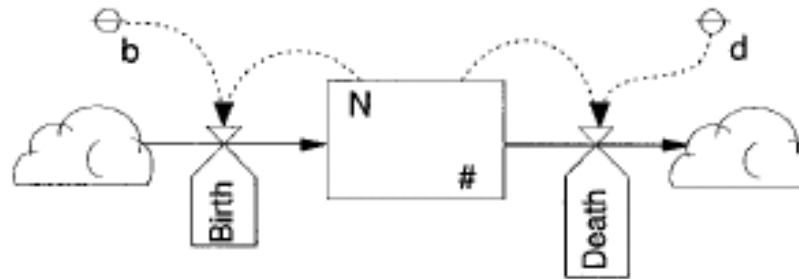


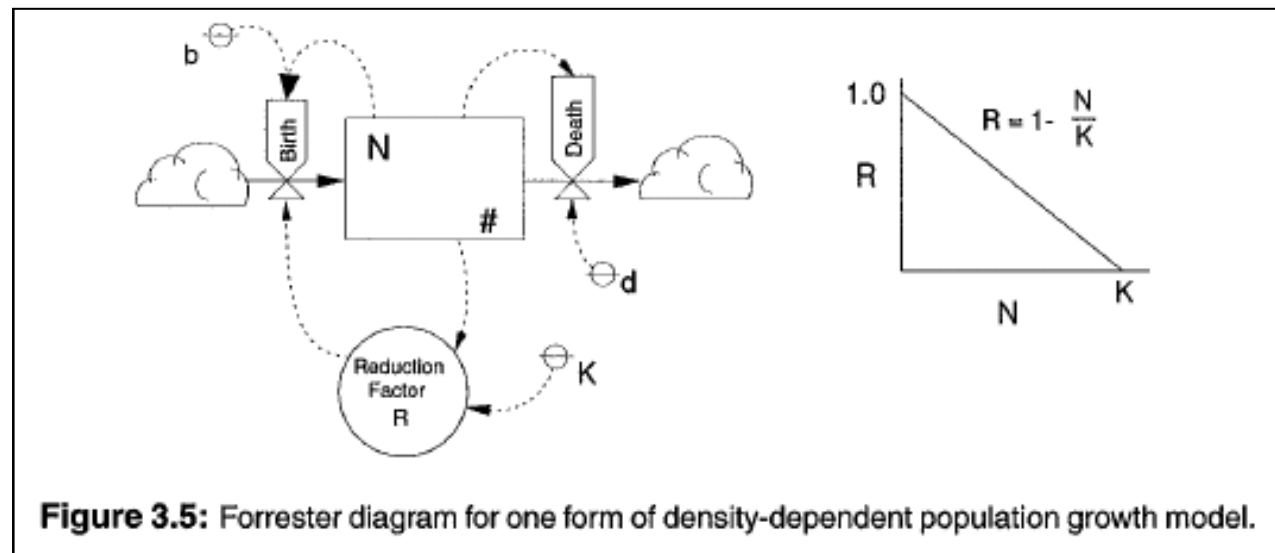
Figure 3.4: Forrester diagram for one form of the density-independent population growth model.



3.3 EXAMPLES

Population Growth with Explicit Birth and Death *Density-Dependent Model*

$$N_{t+1} = N_t + \underbrace{bN_t\left(1 - \frac{N_t}{K}\right)}_R - dN_t \quad (3.2)$$



3.3 EXAMPLES

◎ Net Population Growth

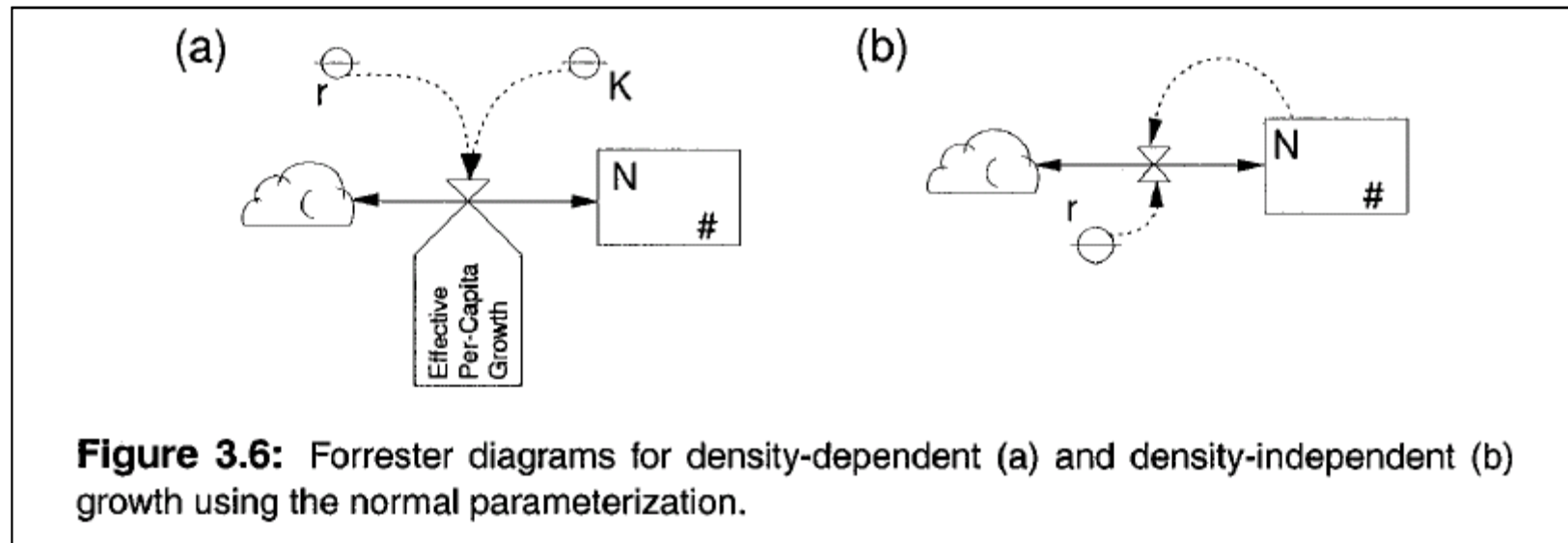


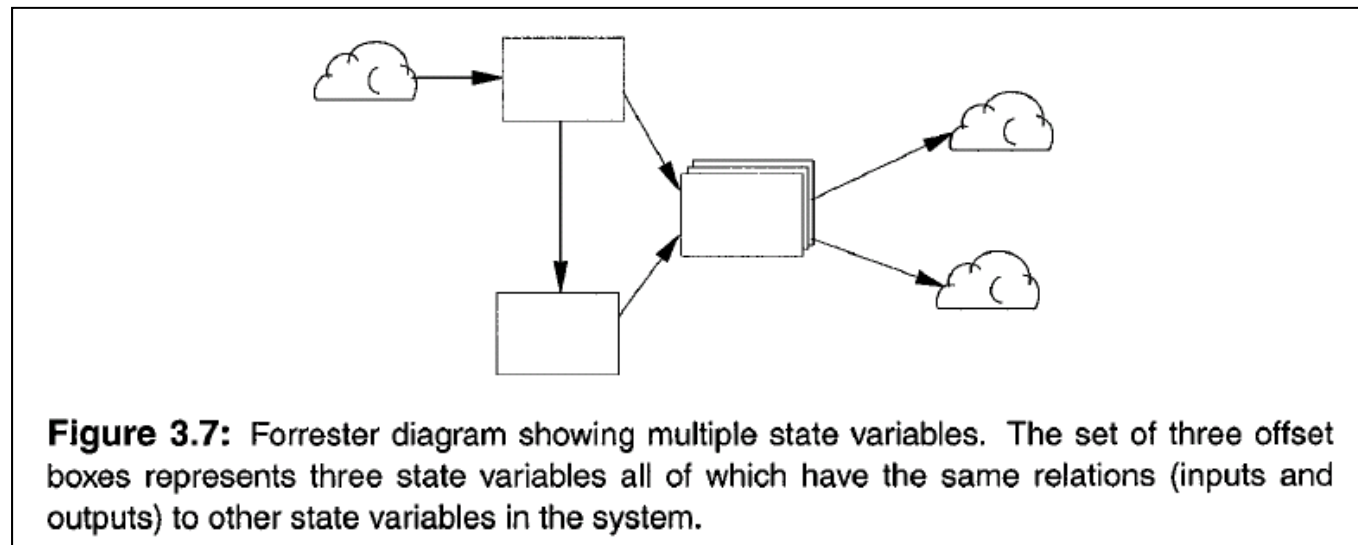
Figure 3.6: Forrester diagrams for density-dependent (a) and density-independent (b) growth using the normal parameterization.



3.3 EXAMPLES

Multiple State Variables

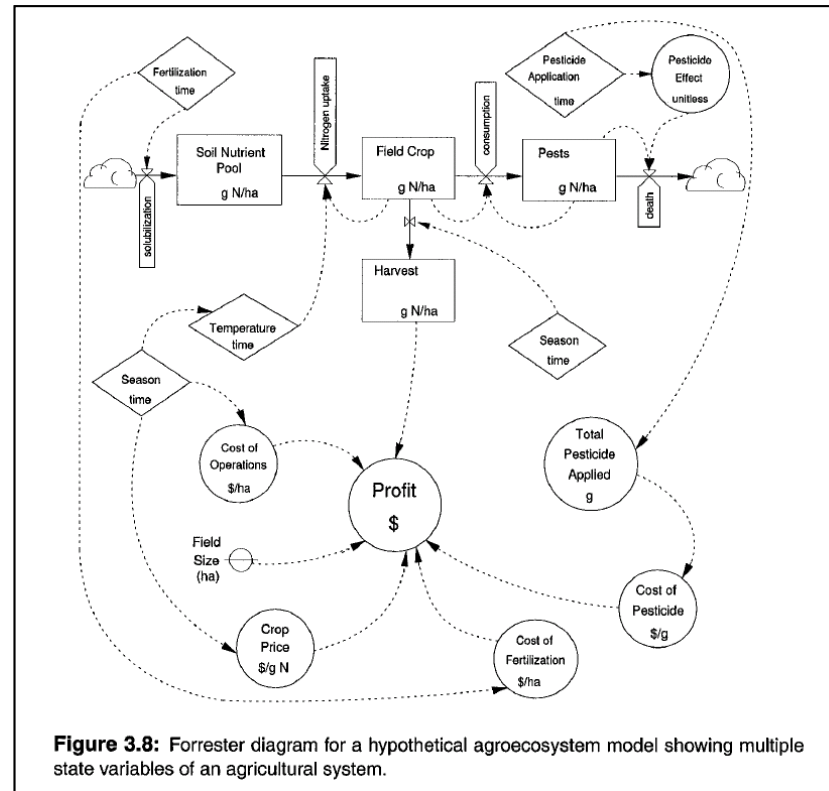
- When a model has more than one state variable, then each object is represented by a box (level) that connects with the others according to the flow of material (energy) defined by the relations (i.e., foraging relationships).



3.3 EXAMPLES

Multiple State Variables

A Hypothetical Agroecosystem Model



3.3 EXAMPLES

- ◎ Multiple Flow Variables and Units
 - ◎ When different units on flow variables are modeled (e.g., g N and g C), parallel models (or multiple models) must be used to avoid having "apples" flow into "oranges."
 - ◎ The dynamics of many biological processes depend on several interacting variables.
 - ◎ Two variables (A and B) are on the same level of biological organization if all of the measurements that can logically be made on A can also be made on B, and there are no measurements that can be made on B that cannot be made on A.



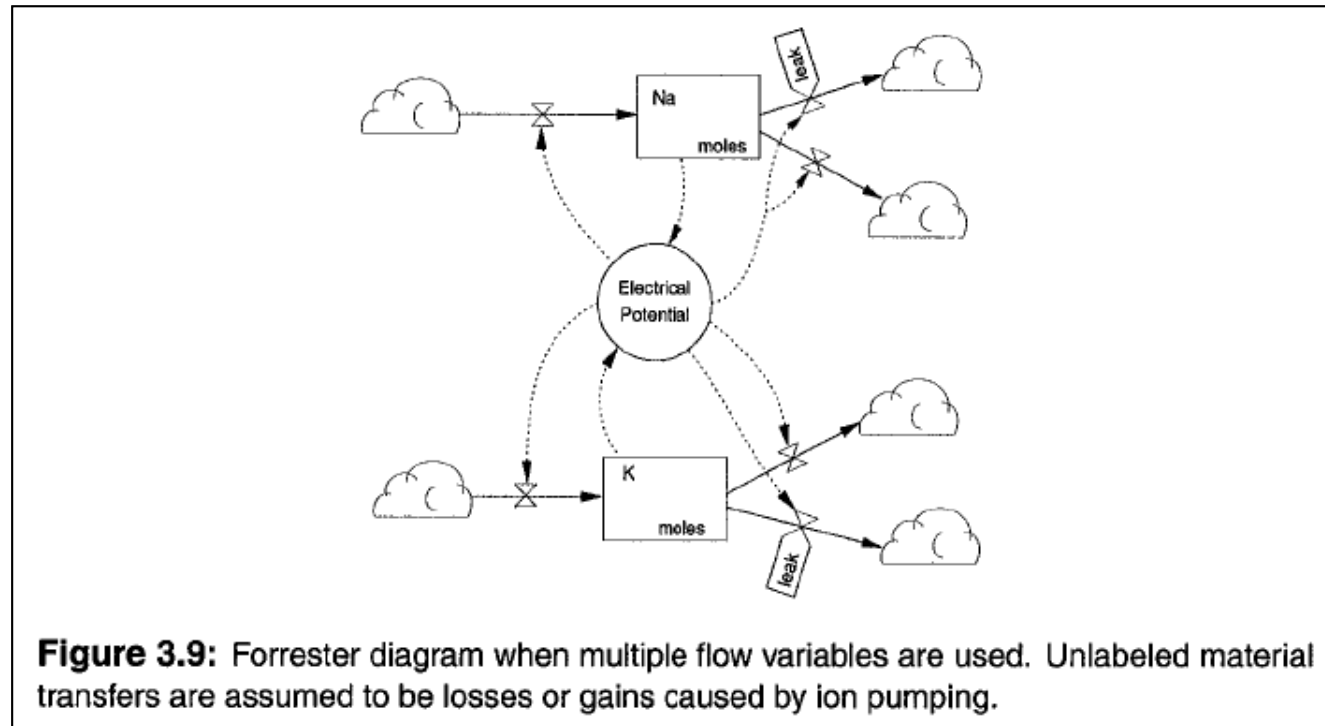
3.3 EXAMPLES

- ◎ Multiple Flow Variables and Units
 - ◎ Variables that are on the same level of organization may interact to affect some biological process negatively (negative feedback), positively (synergism), or independently (substitutable).
 - ◎ If we wish to describe the dynamics of the affected process as influenced by the variables, then we must describe the dynamics of the individual variables and their effect on the process.
 - ◎ Since the physical quantities cannot flow among themselves, we represent the separate dynamics as parallel models.



3.3 EXAMPLES

Multiple Flow Variables and Units *A Simple Model of Nerve Cell Activity*



3.3 EXAMPLES

Multiple Flow Variables and Units *A Simple Predator-Prey Model*

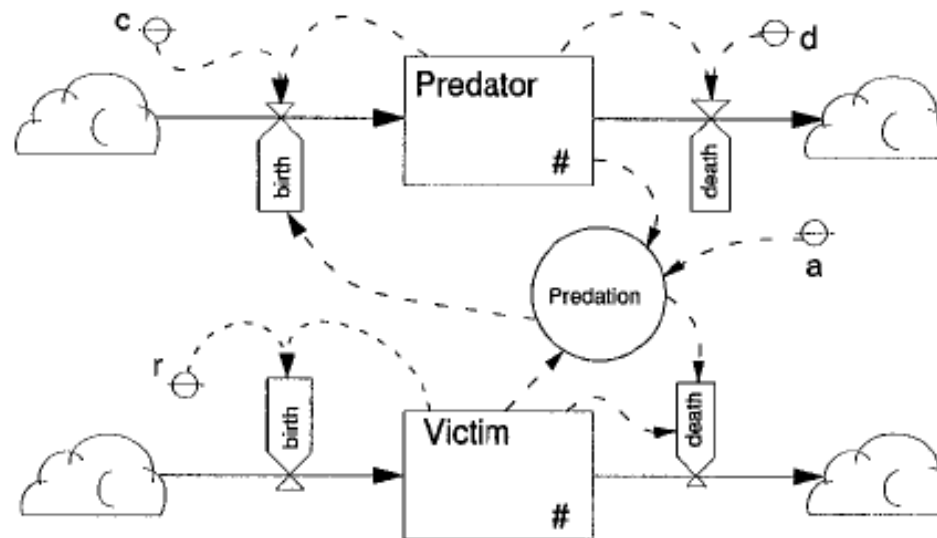


Figure 3.10: Simplified Forrester diagram for linked population models based on numbers of individuals.



3.4 ERRORS IN FORRESTER DIAGRAMS

1. Using any symbols other than those defined in Fig. 3.2. For example, there is no symbol like a solid line with no arrowhead attached (Fig. 3.11a).
2. Failing to label all boxes, variables (auxiliary and driving), and parameters with names and units (where appropriate, Fig. 3.11a).
3. Showing sources or sinks influencing rates (Fig. 3.11b).
4. Showing rates influencing state variables (Fig. 3.11c).
5. Showing information flows directly into state variables (Fig. 3.11a). State variables only change by a change in rates.
6. Showing material flows (solid arrows) between objects other than state variables and sources and sinks (Fig. 3.11d).
7. Showing an influence on a quantity that cannot change (e.g., a parameter, Fig. 3.11d).
8. Showing information flows between state variables (Fig. 3.11e).
9. Using incompatible units of flows or state variables (Fig. 3.11f).
10. Using state variables that are not in the model (objectives or equations) or not including state variables that are in the model.



3.4 ERRORS IN FORRESTER DIAGRAMS

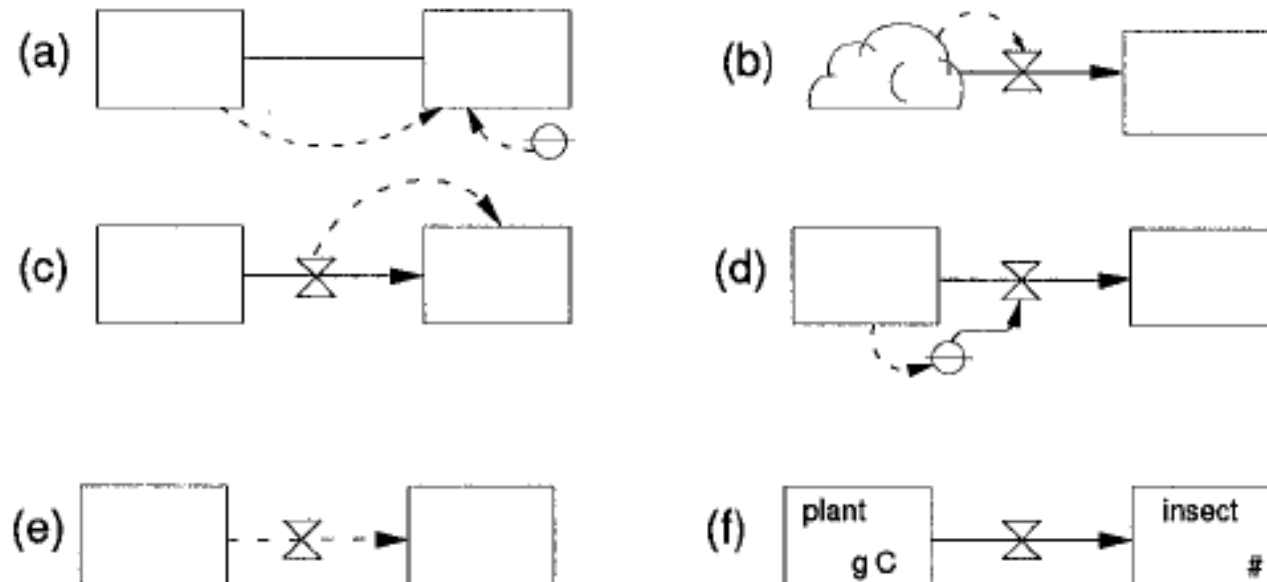


Figure 3.11: Examples of incorrect Forrester diagram fragments.



3.5 ADVANTAGES AND DISADVANTAGES OF FORRESTER DIAGRAM

⊙ Advantages

- ⊙ Forrester diagrams help in learning the rudiments of the system objects.
- ⊙ Understanding is more quickly attained, and constructive criticism is more readily achieved.
- ⊙ Forrester diagrams can be a valuable aid in organizing the computer simulation program.

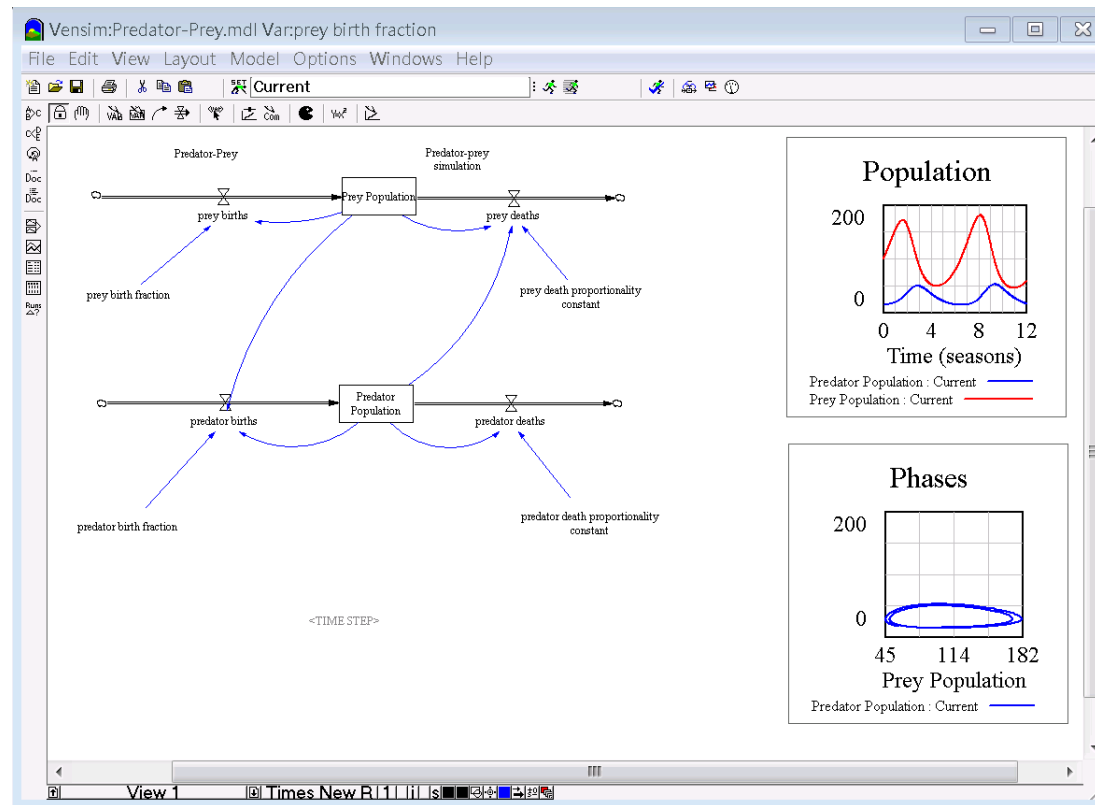
⊙ Disadvantages

- ⊙ There is a point at which diagram complexity obfuscates the basic structure of the model and frustrates attempts to effectively communicate.



3.5 ADVANTAGES AND DISADVANTAGES OF FORRESTER DIAGRAM

◎ Forrester diagrams - Computer simulation program



<http://www.vensim.com/>



3.5 ADVANTAGES AND DISADVANTAGES OF FORRESTER DIAGRAM

◎ Forrester diagrams – Group practice

Draw the Forrester diagram of ***Pan Water Cycle***

◎ ***What does this Model Represent?***

This model represents a model of a pan filled with water. The water will eventually evaporate, at which point it will then condense on the lid of the pan and fall back into the water below. Models such as this can help to simulate the much larger pan of water that is our oceans, whose evaporation yields rain.

◎ ***What are the elements of this Model?***

The water in this model is divided into three categories - water in the pan, water in the air, and water on the cover of the pan. Water can transform between each of these categories through evaporation, condensation, and precipitation. The quantity of water in each area is displayed on the graph to the right.

◎ ***How do I use this Model?***

Set the parameters of the model to your liking to determine the rate at which water evaporates, leaks, condenses, and so on. Then, run the model and view your results on the right.

◎ ***What should I expect?***

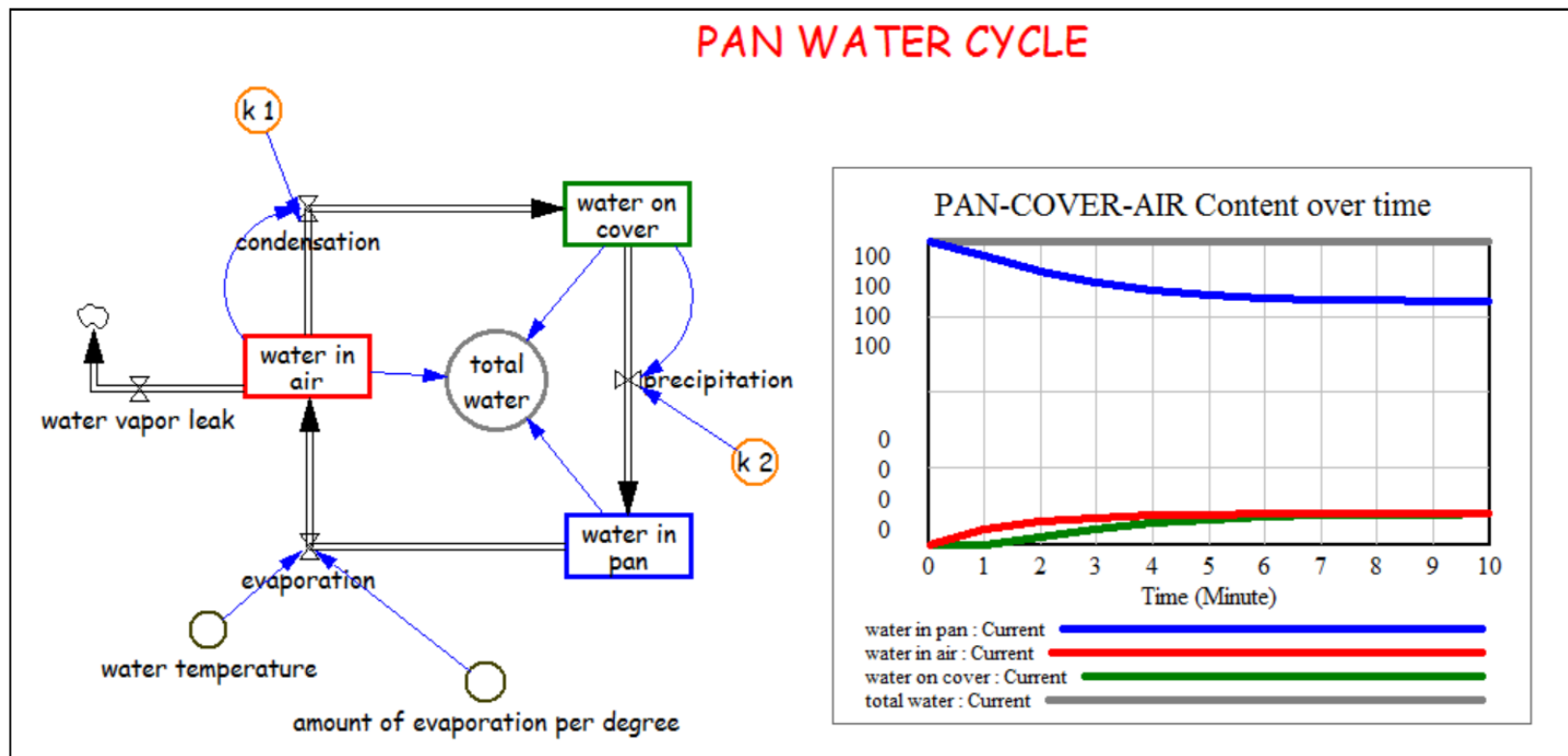
In general, any setup should reach equilibrium at some point, when the water in the pan, in the air, and on the cover is relatively constant. However, the total water will continuously decrease due to leakage.



3.5 ADVANTAGES AND DISADVANTAGES OF FORRESTER DIAGRAM

◎ Forrester diagrams – Group practice

Draw the Forrester diagram of *Pan Water Cycle*



3.6 PRINCIPLES OF QUALITATIVE FORMULATION

◎ Key Steps in Qualitative Formulation

- Identify *the state variables* (levels)
- Identify the *flows* among the state variables
- Identify the *controls* on the flow rates
- Identify the *auxiliary* and *driving variables*



3.6 PRINCIPLES OF QUALITATIVE FORMULATION

☉ Questions Need to be Answered:

- What are the questions to be answered?
- What quantities are needed to answer the questions?
- What equations will answer the questions?
- What other primary flow quantities are needed?
- Is an explicit spatial representation required?
- What are the controls on the flow rates between the state variables?
- Do you know any parameter names?



3.7 MODEL SIMPLIFICATION

Eliminate State Variables

- Convert a state variable into a constant or an auxiliary variable.
- Aggregate state variables.

Make "Stronger" Assumptions

- Convert functions of state variables into constants.
- Convert nonlinear relationships into linear relationships.

Remove Temporal Complexity

- Convert random models into deterministic models.
- Convert driving variables to constants.

Remove Spatial Complexity



3.8 OTHER MODELING PROBLEMS

◎ Transport Models

- ◎ In transport models, we have a substance [energy (heat) or a quantity of matter] that flows from spatial point to point.
- ◎ Conceptual rate equation:

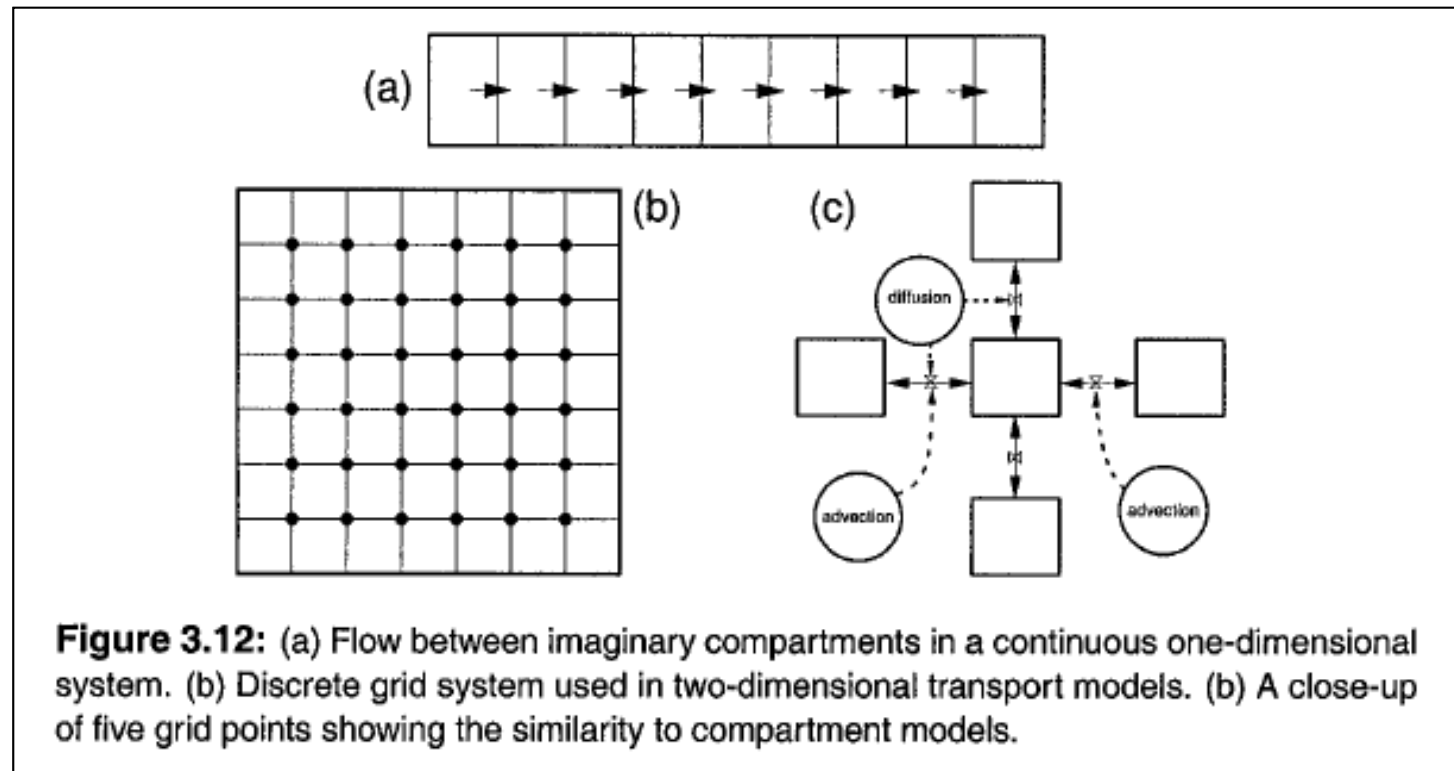
$$\frac{\partial p(x,t)}{\partial t} = \left(\begin{matrix} \text{Advection} \\ \text{In} \end{matrix} \right) - \left(\begin{matrix} \text{Advection} \\ \text{Out} \end{matrix} \right) + \left(\begin{matrix} \text{Diffusion} \\ \text{In} \end{matrix} \right) - \left(\begin{matrix} \text{Diffusion} \\ \text{Out} \end{matrix} \right) + \left(\begin{matrix} \text{Pollutant} \\ \text{Creation} \end{matrix} \right) - \left(\begin{matrix} \text{Pollutant} \\ \text{Distruction} \end{matrix} \right)$$

- ◎ A compartment model paradigm and the Forrester diagram approach are not always appropriate for transport model when the system is modeled as spatially continuous with small spatial resolution. Nevertheless, at least in early model formulation stages, the compartment model concept can be useful for transport models.



3.8 OTHER MODELING PROBLEMS

◎ Transport Models



3.8 OTHER MODELING PROBLEMS

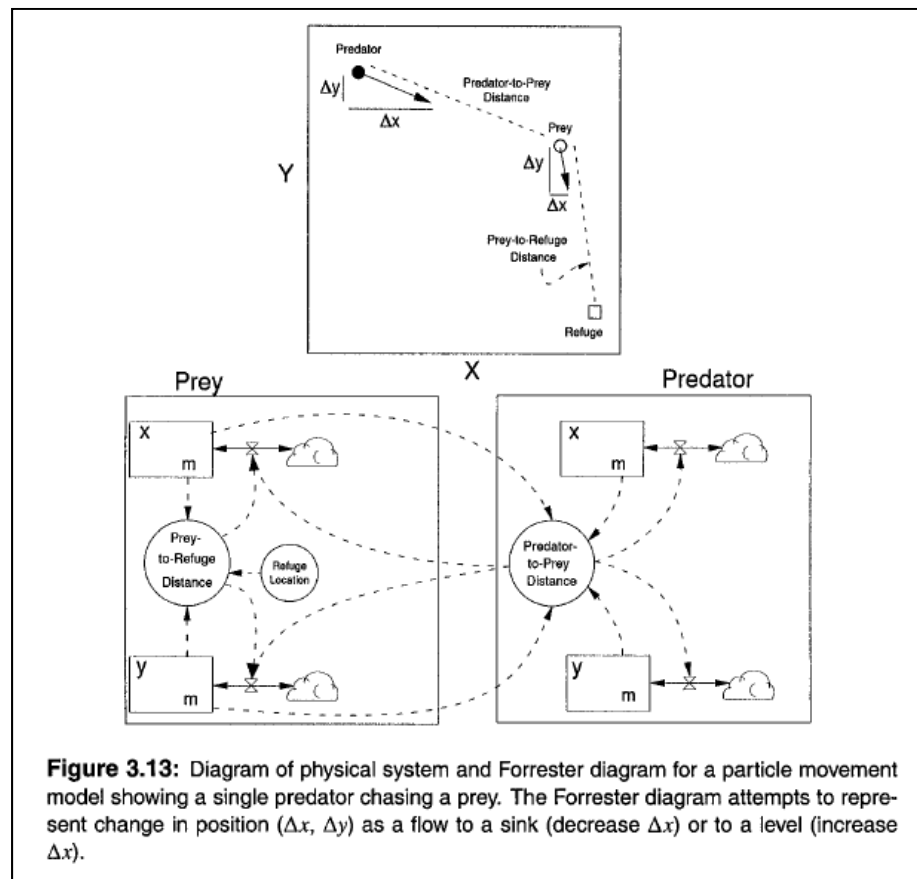
◎ Particle Models

- ◎ Particle models describe systems in which the variables are physical objects (e.g., billiard balls, or individual organisms) that change in some way according to dynamic equations. This is called the *Lagrangian* frame of reference, as opposed to the *Eulerian* approach of transport models.
- ◎ As with the transport model, Forrester diagrams can be useful for initial model formulation and detailing a subset of the objects and interactions. But it is not useful to describe all of the objects this way.



3.8 OTHER MODELING PROBLEMS

Particle Models



3.8 OTHER MODELING PROBLEMS

Finite State Models

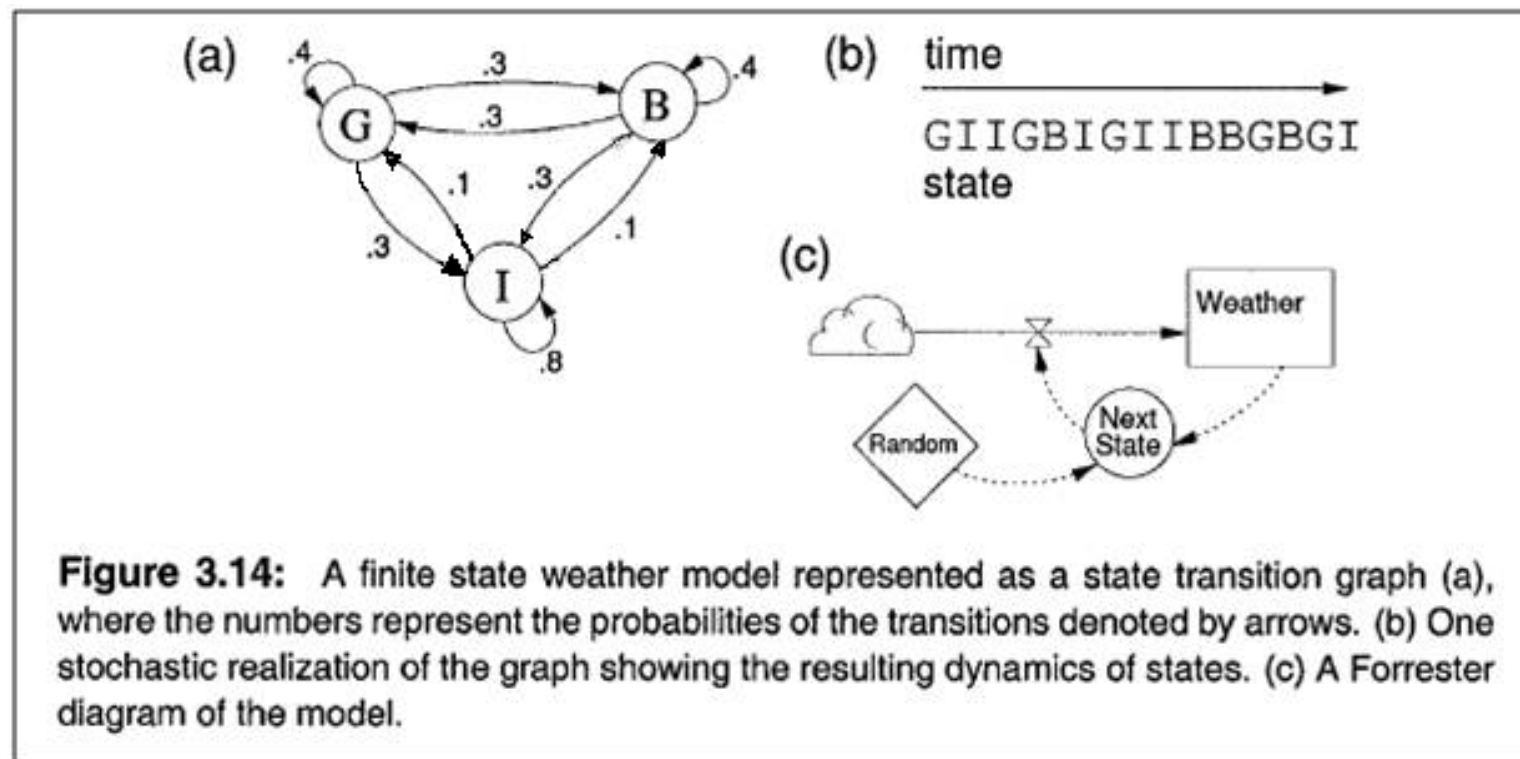


Figure 3.14: A finite state weather model represented as a state transition graph (a), where the numbers represent the probabilities of the transitions denoted by arrows. (b) One stochastic realization of the graph showing the resulting dynamics of states. (c) A Forrester diagram of the model.

